Stars and Bars

Stars and Bars answers distribution questions.

How many ways can we distribute identical items into distinct boxes under some constraints?

Key words you'll see in CP:

- distribute
- split
- partition
- nonnegative integers
- sum equals something

Problem 1 (Base Case)

In how many ways can we distribute n identical balls into k distinct boxes, allowing empty boxes?

Translate into math

Let:

x1 + x2 + x3 + ... + xk = n

where:

xi ≥ 0 (can be zero)

This is the core Stars and Bars problem.

Visual (Stars and Bars Idea)

Step 1: Represent balls as stars

Example: n = 5

* * * * *

Step 2: Separate into k boxes using bars

We need k-1 bars.

Example: k = 3

* * | * * * | *

Interpretation:

Box 1 → 2 stars
Box 2 → 3 stars
Box 3 → 1 star

Step 3: What are we really counting?

We are arranging:

- n stars
- k-1 bars

Total symbols:

n + k - 1

We just need to choose positions for the bars.

Stars and Bars Formula

Number of ways:

$$\binom{n+k-1}{k-1}$$

(or equivalently)

$$\binom{n+k-1}{n}$$

Why?

Total positions = n + k - 1
Choose k - 1 positions for bars

Example

Number of non-negative integer solutions to:

x1 + x2 + x3 = 7

Apply formula

$$\binom{7+3-1}{3-1} = \binom{9}{2} = 36$$

CP Interpretation Patterns

Whenever you see:

x1 + x2 + ... + xk = n

xi ≥ 0

→ Stars and Bars

No Empty Boxes Case (xi ≥ 1)

Problem

Distribute n identical balls into k distinct boxes, each box must get at least one ball

Step 1: Guarantee one ball per box

Let:

yi = xi - 1

Now:

y1 + y2 + ... + yk = n - k

with:

yi >= 0

Step 2: Apply stars and bars

$$\binom{(n-k)+k-1}{k-1} = \binom{n-1}{k-1}$$

Example

x1 + x2 + x3 = 7, xi ≥ 1

$$\binom{7-1}{3-1} = \binom{6}{2} = 15$$

Lower Bounds Other Than 0 or 1

Problem

x1 + x2 + x3 = 20

with:

x1 ≥ 2, x2 ≥ 5, x3 ≥ 3

Step 1: Shift variables

Let:

y1 = x1 - 2
y2 = x2 - 5
y3 = x3 - 3

Now:

y1 + y2 + y3 = 20 - (2 + 5 + 3) = 10

and

yi ≥ 0

Step 2: Apply stars and bars

$$\binom{10+3-1}{3-1} = \binom{12}{2}$$

Upper Bounds

Stars and Bars does NOT directly handle upper bounds.

Example:

x1 + x2 + x3 = 10
0 ≤ xi ≤ 5

Solution approach:

Use Inclusion–Exclusion Principle